SB8 · MODAL-BEAM

Deep simply-supported beam — shear-flexible (Timoshenko) natural frequencies

Timoshenko frame shear deformation — deep-beam natural frequencies vs the exact Timoshenko continuum

PASS

error +0.0478% · tol 0.05%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine 2-node Timoshenko frame element** — shear-flexible
- Rectangular shear area **Avy=Avz=5/6·A** (calcShearArea)
- Deep beam **L/h = 5** — shear term dominates
- Lumped mass via **Loads-to-Masses** (MY)
- Six modes vs **exact Timoshenko continuum**
- Shear knockdown **4%→52%** below Euler

§3 PROBLEM DESCRIPTION

STRIX models frame members as **shear-flexible (Timoshenko) beams** carrying the real shear areas (rectangular `Avy=Avz=5/6·A`). The seismic benchmarks (SM5, SM6, SR1, SR2b) use that element on **slender** members, where the shear term contributes a fraction of a percent — none verifies the shear contribution in isolation. SB8 fills that gap. A **deep** simply-supported beam (span/depth `L/h=5`, square section so `Iy=Iz` and the vibration plane is immaterial) is discretised into STRIX engine frame elements with **lumped translational mass** injected through the production Loads-to-Masses path (downward `Fz=−m·g`, direction `Y` → `my=m`) so the lowest six modes are the sequential horizontal bending modes. Shear deformation lowers the natural frequencies from **4.4% (mode 1) to 52.4% (mode 6)** below the Euler-Bernoulli values, and STRIX must reproduce the **shear-lowered** frequencies. The feature is verified through the modal path, where the shear effect grows with mode order, against the **exact continuum** simply-supported Timoshenko frequencies (a closed form with no root-finding); the coarse→fine sweep demonstrates convergence.

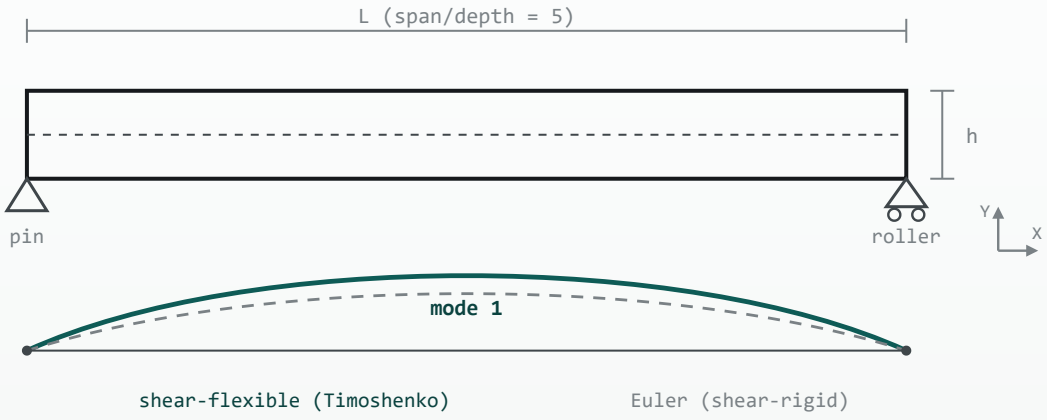


Figure SB8. Deep simply-supported beam (span/depth = 5) with the first bending mode. The shear-flexible (Timoshenko) shape vibrates at a measurably lower frequency than the Euler (shear-rigid) beam — the effect grows with mode order.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		SECTION & MATERIAL		MASS & SOLVER	
Span L	3000 mm	E	30000 MPa	Density	2400 kg/m³
Section	600 × 600 mm (square)	ν	0.2	Injection	Loads-to-Masses (MY)
Span / depth	5	G	12500 MPa	Finest mesh	256 elem
Inertia I	1.08×10¹⁰ mm⁴	Shear area	5/6·A (Timoshenko)	Solver	STRIX engine (eigen)

§5 REFERENCE SOLUTION

Natural frequency f_1: $f_n=(1/2\pi)\sqrt{\omega_n^2}$, $\omega_n^2=(EI\beta_n^4/\rho A)/(1+EI\beta_n^2/\kappa GA)$, $\beta_n=n\pi/L$ (SS Timoshenko, no rotary inertia) = 102.149414 Hz

Natural frequency f_2: $\omega_2^2=(EI\beta_2^4/\rho A)/(1+EI\beta_2^2/\kappa GA)$, $\beta_2=2\pi/L$ = 364.059185 Hz

Natural frequency f_3: $\omega_3^2=(EI\beta_3^4/\rho A)/(1+EI\beta_3^2/\kappa GA)$, $\beta_3=3\pi/L$ = 706.690082 Hz

Natural frequency f_4: $\omega_4^2=(EI\beta_4^4/\rho A)/(1+EI\beta_4^2/\kappa GA)$, $\beta_4=4\pi/L$ = 1078.102736 Hz

Natural frequency f_5: $\omega_5^2=(EI\beta_5^4/\rho A)/(1+EI\beta_5^2/\kappa GA)$, $\beta_5=5\pi/L$ = 1455.799333 Hz

Natural frequency f_6: $\omega_6^2=(EI\beta_6^4/\rho A)/(1+EI\beta_6^2/\kappa GA)$, $\beta_6=6\pi/L$ = 1832.019784 Hz

Exact continuum reference: simply-supported Timoshenko (shear-flexible) beam with translational mass and rotary inertia excluded — matching the production lumped translational-mass model. The SS frequencies have a closed form with no root-finding, $\omega_n^2 = (EI\cdot\beta_n^4/\rho A)/(1 + EI\cdot\beta_n^2/\kappa GA)$ with $\beta_n=n\pi/L$, which reduces to the exact Euler SS frequency $\omega E_{,n}^2=EI\cdot\beta_n^4/\rho A$ in the shear-rigid limit $\kappa GA\to\infty$. Reproduced from scratch by sb8Theory() to the values below. The shear term lowers the frequencies by 4.4% (mode 1) to 52.4% (mode 6) below Euler, so STRIX must reproduce the shear-lowered values. Genre context: NAFEMS 'Selected Benchmarks for Natural Frequency Analysis' (deep simply-supported beam, FV5); shear coefficient κ=5/6 (Cowper 1966, rectangular); Timoshenko beam dynamics closed forms (Timoshenko & Young; Blevins, Formulas for Natural Frequency and Mode Shape; Han, Benaroya & Wei 1999).

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Natural frequency f_1 (Hz)	Euler −4.4%	102.145904	102.149414	-0.0034%	PASS
Natural frequency f_2 (Hz)	Euler −14.8%	364.017451	364.059185	-0.0115%	PASS
Natural frequency f_3 (Hz)	Euler −26.5%	706.544024	706.690082	-0.0207%	PASS
Natural frequency f_4 (Hz)	Euler −37.0%	1077.782925	1078.102736	-0.0297%	PASS
Natural frequency f_5 (Hz)	Euler −45.5%	1455.237299	1455.799333	-0.0386%	PASS
Natural frequency f_6 (Hz)	Euler −52.4%	1831.143749	1832.019784	-0.0478%	PASS

§7 MESH CONVERGENCE (FUNDAMENTAL FREQUENCY)

MESH	F_1 (HZ)	DIFF
4 elem	101.8875	−0.2564%
8 elem	102.0906	−0.0576%
16 elem	102.1351	−0.014%
32 elem	102.1459	−0.0034%
64 elem	102.1483	−0.0011%
128 elem	102.1503	+0.00090%
256 elem	102.1459	−0.0034%

The Timoshenko frame element is exact for the shear-flexible static beam, so the discretisation error here comes mainly from **lumped mass**. The fundamental frequency rises from −0.26% (4 elements) to within 0.004% of the closed form by 32 elements; finer meshes stay within that band (+0.0009% at 128, −0.0034% at 256). At the finest mesh (256 elements) STRIX reproduces the exact Timoshenko continuum frequencies to `under 0.05% across all six modes` (worst 0.048%, mode 6; §6) — and lands on the shear-lowered values, **not** the Euler ones, confirming the shear-deformation term is active and correctly sized.

§8 CONCLUSION

PASS

STRIX reproduces the **exact continuum simply-supported Timoshenko frequencies** of a deep (L/h=5) beam to **within 0.05%** across the first **six bending modes**, landing on the shear-lowered values (**4.4% to 52.4% below Euler**) rather than the Euler-Bernoulli ones. This isolates and verifies the **shear-deformation term** of the production **Timoshenko frame** element — the element every eigenvalue and response-spectrum analysis relies on, whose shear contribution had until now only been exercised implicitly on slender members. SB8 closes that gap in the modal foundation of the seismic pipeline.

§9 REFERENCES & ANALYSIS FILES

1. NAFEMS — *Selected Benchmarks for Natural Frequency Analysis.* Deep simply-supported beam (free-vibration benchmark FV5); genre context for shear-flexible beam natural frequencies.
2. Cowper, G. R. — "The Shear Coefficient in Timoshenko's Beam Theory," *J. Applied Mechanics*, 1966 (rectangular section κ = 5/6).
3. Timoshenko, S. & Young, D. H. — *Vibration Problems in Engineering;* Blevins, R. D. — *Formulas for Natural Frequency and Mode Shape;* Han, Benaroya & Wei, 1999 (closed-form Timoshenko beam frequencies).
4. STRIX headless harness `buildEigenPy (Loads-to-Masses MY) → STRIX engine (eigen route) → modal_props.txt`; 2-node Timoshenko frame element with shear areas Avy/Avz; reference `sb8Theory()` (exact SS Timoshenko closed form).

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SB8/` `sha256 (pending publish)`